Project Timeline

10-week schedule, milestones, dependencies and critical path

Document	07 - Project Timeline
Project	AI Customer Support Platform (fictional company: Kestrel Commerce Ltd.)
Prepared by	Project Manager (portfolio author) - self-directed simulation
Status	Baseline v1.0
Project window	5 Jan - 13 Mar 2026 (simulated project window)

SIMULATED PROJECT - Self-directed portfolio case study. Not professional experience, not client work, not employment. All people, figures and outcomes are fictional.

Planned 10-week schedule (Mon 5 Jan - Fri 13 Mar 2026) built from the WBS work packages. Durations are working days (Mon-Fri, no public holidays assumed). Dependencies: FS = finish-to-start, SS = start-to-start. The visual schedule is in 08_Gantt_Chart.xlsx (and the PNG/PDF preview). Task IDs match the WBS; the 'Stories' column links each task to the backlog.

1. Sprint calendar

Sprint	Dates	Weeks	Theme	Points	Review / demo
Sprint 1	5 Jan - 16 Jan 2026	W1-W2	Foundation & Discovery	28	16 Jan 2026
Sprint 2	19 Jan - 30 Jan 2026	W3-W4	Knowledge Base & AI Core	36	30 Jan 2026
Sprint 3	2 Feb - 13 Feb 2026	W5-W6	Customer Chat Experience	34	13 Feb 2026
Sprint 4	16 Feb - 27 Feb 2026	W7-W8	Human Handover, Analytics & Hardening	34	27 Feb 2026
Sprint 5	2 Mar - 13 Mar 2026	W9-W10	Readiness, Release & Closure	29	13 Mar 2026

2. Milestones

ID	Milestone	Date	Task	Evidence of completion
M1	Project Charter approved	7 Jan 2026	1.1	Approved Project Charter; Stakeholder Register v1
M2	Requirements & UX baseline approved	16 Jan 2026	2.2	Requirements v1.0 baseline; backlog estimated
M3	Sprint 2 review: KB ingestion + RAG answers demo	30 Jan 2026	5.3	RAG answer API; fallback and out-of-scope handling
M4	Sprint 3 review: customer chat + escalation request demo	13 Feb 2026	8.2	Integrated chat flow in staging; demo
M5	Feature complete / code freeze	27 Feb 2026	8.3	Feature-complete build in staging
M6	UAT sign-off	6 Mar 2026	9.6	UAT results; defect fixes; written UAT sign-off
M7	Go / No-Go decision	9 Mar 2026	10.3	Signed Go/No-Go checklist
M8	Pilot release to production	11 Mar 2026	10.4	Production deployment; smoke-test results
M9	Project closure	13 Mar 2026	12.2	Closure Report; Lessons Learned

3. Task schedule (ordered by start date)

ID	Task	Start	End	Days	Dependency	Responsible role	Deliverable	Stories	Milestone
1.1	Kickoff, Project Charter & stakeholder register	5 Jan	7 Jan	3	-	Project Manager	Approved Project Charter; Stakeholder Register v1	-	M1
1.2	Project planning (scope, WBS, schedule, risks, RACI, sprint plan)	8 Jan	14 Jan	5	1.1 (FS)	Project Manager	Scope, WBS, timeline, Risk Register, RACI, Sprint Plan baseline	-	-
1.3	Monitoring & controlling (ceremonies, risk reviews, status reporting, change control)	8 Jan	13 Mar	47	1.1 (FS)	Project Manager	Sprint reviews, status report, decision and change logs	-	-
2.1	Requirements elicitation (interviews, ticket-category analysis)	8 Jan	13 Jan	4	1.1 (FS)	Product Manager	Interview notes; ticket-category analysis	-	-
3.1	UX research & user flows	8 Jan	13 Jan	4	1.1 (FS)	UI/UX Designer	User flows (chat, escalation, admin)	-	-
5.1	LLM & embedding evaluation spike	8 Jan	16 Jan	7	1.1 (FS)	AI/ML Engineer	Spike report; provider recommendation + fallback option	US-11	-
10.1	Environments & CI/CD pipeline	8 Jan	16 Jan	7	1.1 (FS)	DevOps Engineer	Dev/staging/prod environments; CI/CD pipeline	US-01	-
2.2	Requirements documentation & baseline	12 Jan	16 Jan	5	2.1 (SS)	Product Manager	Requirements v1.0 baseline; backlog estimated	-	M2
3.2	Wireframes & clickable prototype	12 Jan	16 Jan	5	3.1 (SS)	UI/UX Designer	Clickable prototype for all MVP screens	US-03	-
4.1	Staff authentication & role-based access	12 Jan	16 Jan	5	6.1 (SS)	Backend Developer	Staff login; RBAC permission model	US-04, US-05	-
6.1	Data model & schema	12 Jan	16 Jan	5	2.1 (SS)	Backend Developer	Schema v1 and migrations; backup configuration	US-02	-
6.3	KB content audit & cleanup (with Support SMEs)	12 Jan	30 Jan	15	2.1 (SS)	Product Manager	Audited KB: duplicates removed, outdated articles fixed	-	-
3.3	Usability check with support agents & design handoff	19 Jan	23 Jan	5	3.2 (FS)	UI/UX Designer	Usability findings; design handoff package	-	-
4.2	KB management APIs (CRUD, publish, bulk import)	19 Jan	28 Jan	8	4.1 (FS)	Backend Developer	KB APIs; import validation	US-09, US-10	-
5.2	KB ingestion pipeline (chunk, embed, index)	19 Jan	27 Jan	7	5.1 (FS), 6.1 (FS)	AI/ML Engineer	Ingestion pipeline; re-index job	US-12	-
6.2	Vector store setup & seed KB load	19 Jan	23 Jan	5	6.1 (FS), 5.1 (FS)	AI/ML Engineer	Vector index ready; cleaned KB loaded	US-12	-
7.1	Frontend scaffold & Admin KB UI	19 Jan	30 Jan	10	3.2 (FS)	Frontend Developer	App scaffold; admin KB list, editor and import screens	US-09, US-10	-

ID	Task	Start	End	Days	Dependency	Responsible role	Deliverable	Stories	Milestone
9.1	Test strategy & golden Q&A set	19 Jan	30 Jan	10	2.2 (FS)	QA Engineer	Test strategy; golden set (100+ questions); evaluation script	US-15	-
5.3	RAG answer service with citations, guardrails & fallback	22 Jan	30 Jan	7	5.2 (SS)	AI/ML Engineer	RAG answer API; fallback and out-of-scope handling	US-13, US-14	M3
4.3	Customer sign-in (email one-time code)	2 Feb	6 Feb	5	4.1 (FS)	Backend Developer	Customer OTP sign-in	US-06	-
4.4	Chat, conversation-history & escalation-request APIs	2 Feb	11 Feb	8	5.3 (FS)	Backend Developer	Chat, history and escalation-request APIs	US-18, US-19	-
5.4	PII redaction & answer-quality tuning	2 Feb	11 Feb	8	5.3 (FS)	AI/ML Engineer	PII redaction layer; tuned prompts and thresholds	US-25	-
7.2	Customer chat UI (chat, feedback, history, 'Talk to a person')	2 Feb	13 Feb	10	3.3 (FS), 5.3 (FS)	Frontend Developer	Customer chat experience	US-16, US-17, US-18, US-19	-
8.1	Email provider integration (OTP & notifications)	2 Feb	4 Feb	3	10.1 (FS)	Backend Developer	Email integration; SPF/DKIM verified	US-06	-
8.2	End-to-end chat integration (UI - API - RAG - DB)	9 Feb	13 Feb	5	4.4 (SS), 7.2 (SS)	Backend Developer	Integrated chat flow in staging; demo	US-16, US-19	M4
9.2	Sprint QA & AI quality evaluation runs (#1 baseline, #2 post-tuning)	9 Feb	27 Feb	15	9.1 (FS)	QA Engineer	Evaluation reports #1 and #2; regression results	-	-
4.5	Escalation queue, agent reply & notification APIs	16 Feb	24 Feb	7	4.4 (FS), 8.1 (FS)	Backend Developer	Queue, claim, reply, resolve and notification APIs	US-20, US-21, US-22	-
4.6	Event tracking & analytics APIs	16 Feb	19 Feb	4	4.4 (FS)	Backend Developer	Event schema and aggregation queries	US-23	-
4.7	Security hardening	16 Feb	25 Feb	8	4.1 (FS)	Backend Developer	Rate limiting, validation, secrets, dependency scans	US-07	-
7.3	Agent console UI	16 Feb	25 Feb	8	4.5 (SS)	Frontend Developer	Escalation queue and agent console	US-20, US-21	-
7.4	Manager dashboard UI	20 Feb	27 Feb	6	4.6 (FS)	Frontend Developer	Analytics dashboard	US-24	-
9.3	Security testing	23 Feb	27 Feb	5	4.7 (SS)	QA Engineer	Access-control, injection and prompt-injection test results	US-07	-
9.5	UAT preparation	23 Feb	27 Feb	5	9.1 (FS)	QA Engineer	UAT test cases; participants and environment ready	-	-
11.1	User guides & training materials	23 Feb	6 Mar	10	7.3 (SS)	Product Manager	Admin guide; agent guide; 'what the AI can/cannot answer' sheet	US-30	-

ID	Task	Start	End	Days	Dependency	Responsible role	Deliverable	Stories	Milestone
8.3	Full escalation-flow integration (chat - queue - agent - customer)	24 Feb	27 Feb	4	4.5 (SS), 7.3 (SS)	Backend Developer	Feature-complete build in staging	US-22	M5
4.8	Data retention, deletion & audit logging	2 Mar	5 Mar	4	4.5 (FS)	Backend Developer	Retention job; customer data deletion; audit log	US-26, US-08	-
9.4	Performance & load testing	2 Mar	4 Mar	3	8.3 (FS)	QA Engineer	Load test report (50 concurrent sessions)	US-27	-
9.6	UAT execution & defect fixing	2 Mar	6 Mar	5	8.3 (FS), 9.5 (FS)	QA Engineer	UAT results; defect fixes; written UAT sign-off	US-28	M6
10.2	Monitoring, alerts & staging release rehearsal	2 Mar	6 Mar	5	8.3 (FS)	DevOps Engineer	Dashboards and alerts; rehearsed rollback	US-29	-
11.2	Technical documentation & runbook	2 Mar	6 Mar	5	8.3 (FS)	Backend Developer	Architecture and API docs; operations runbook	-	-
10.3	Go / No-Go review	9 Mar	9 Mar	1	9.6 (FS), 9.4 (FS), 10.2 (FS)	Project Manager	Signed Go/No-Go checklist	-	M7
11.3	Support team training session	9 Mar	10 Mar	2	11.1 (FS)	Product Manager	60-minute training delivered; practice tasks completed	US-30	-
10.4	Production pilot release	10 Mar	11 Mar	2	10.3 (FS)	DevOps Engineer	Production deployment; smoke-test results	US-29	M8
10.5	Hypercare & monitoring	11 Mar	13 Mar	3	10.4 (SS)	DevOps Engineer	Daily health check; hot-fix log	-	-
12.1	Final acceptance & handover to Support Operations	11 Mar	12 Mar	2	10.4 (SS)	Project Manager	Sponsor acceptance; handover pack; post-MVP backlog transferred	-	-
12.2	Project Closure Report & lessons learned	12 Mar	13 Mar	2	12.1 (SS)	Project Manager	Closure Report; Lessons Learned	-	M9

4. Critical path and schedule risk

The longest chain of dependent work with the least slack is: 1.1 Charter, 5.1 LLM spike, 5.2 Ingestion, 5.3 RAG service (M3), 4.4 Chat APIs, 4.5 Escalation and reply APIs, 8.3 Full escalation integration (M5), 9.6 UAT (M6), 10.3 Go/No-Go (M7), 10.4 Production release (M8), 12.1-12.2 Closure (M9).

- Slack is thin: about 2 working days (12-13 Feb) between 4.4 and 4.5 and none after 8.3, which is why Sprint 5 holds the Should-Have de-scope buffer and why scope changes after M2 are controlled.
- Highest schedule exposure: the AI core (5.2-5.3) feeds everything in Sprints 3-4; a one-week slip there would move M4 and M5. Early evaluation runs (9.2) and a pre-agreed de-scope order are the response.
- External dependencies on the path: LLM provider terms (D-01) and email domain verification (D-04) - tracked in 18_Dependency_Log.
- Team capacity is 61 person-weeks against about 55 planned (275 of 305 person-days), leaving roughly 10% buffer (see WBS 'Capacity Check').

5. Consistency with other artifacts

- WBS: every task here is a Level-2 work package in 06_Work_Breakdown_Structure.xlsx with the same ID.
- Backlog: each of the 30 stories is linked to at least one task, and every story's sprint matches the sprint in which its linked task ends.
- Sprints: sprint boundaries are Mondays and Fridays; the five reviews fall on 16 Jan, 30 Jan, 13 Feb, 27 Feb and 13 Mar.
- Risks: schedule risks R09, R10, R12 and R13 use the triggers defined against these dates.